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Web Programming Report

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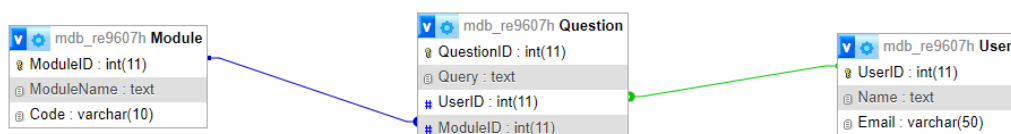
Contents:

Introduction:	1
Design:	1
Technologies Used:	3
Legal, Social and Ethical Issues:	3
Legal issues:	3
Social issues:	5
Ethical considerations:	5
Overview:	6
Testing:	11
Conclusions and future recommendations:	12
References:	12

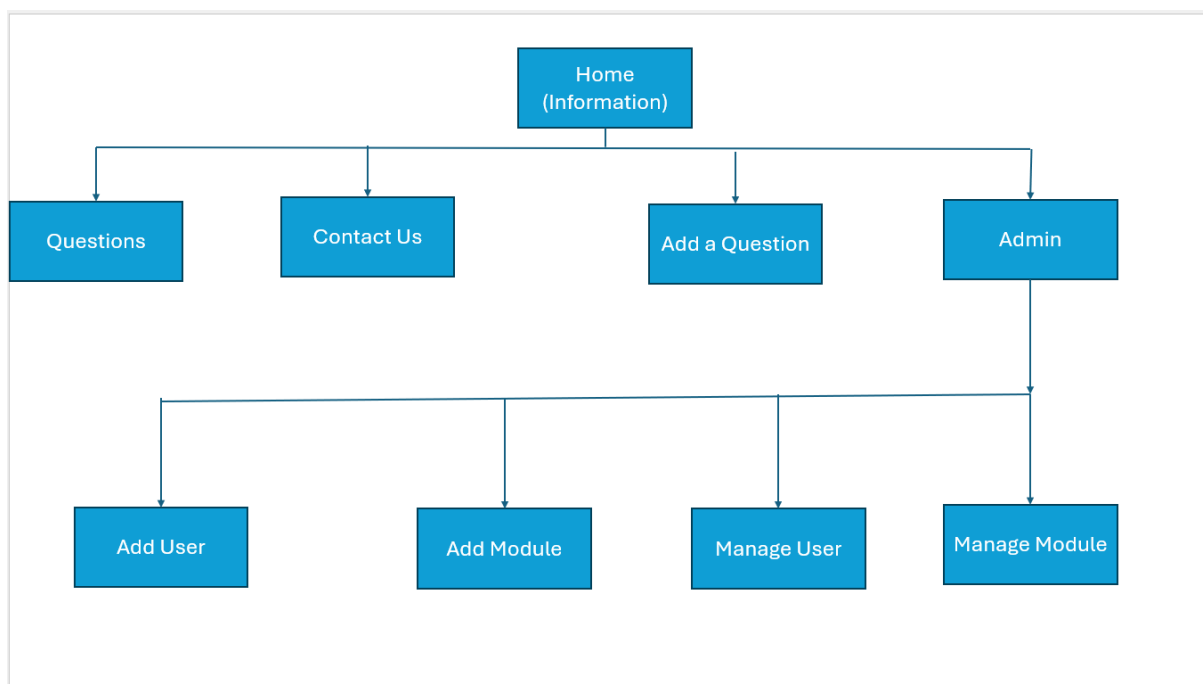
Introduction:

This project involves developing a web-based platform where students may post academic queries and engage with one another. The system will display a list of questions or postings from a database, allowing users to add, amend, and remove them. Students will also be able to include photographs or screenshots with their submissions. Additionally, the site will have a contact form where students may submit messages to the system administrator. Users may manage their profiles by adding, modifying, and removing information associated with their postings. Finally, the system will allow postings to be allocated to certain course modules via dropdown menus.

Design:



The Entity-Relationship Diagram (ERD) for this system has three primary tables: Module, User, and Question, which are the database's fundamental components. The Module table contains information on several modules, including ModuleID and ModuleName, which are used to categorise student questions. The User table provides information about the students, such as UserID, Name, and Email, which connects them to the questions they post. The Question database contains the actual student enquiries, with attributes such as QuestionID, Query, and foreign keys referring both the User (author) and the Module (category). These foreign keys provide links between the UserID in the Question table and the User database, as well as the ModuleID and the Module table, allowing for efficient data retrieval, such as filtering questions by module or user.



The navigation tabs Information, Questions, Post a Question, Contact Us, and Admin are thoughtfully designed to give a simple and straightforward user experience. The Information tab acts as an introduction part, providing important information about the platform and its features. The Questions page allows visitors to explore existing questions and answers, which promotes knowledge exchange. Post a Question allows students to ask their own questions, which promotes involvement. The Contact Us page allows users to easily contact administrators for help or enquiries, ensuring open communication. Finally, the Admin tab is intended for administrators, who may control users, modules and content. The layout ensures each function is simply accessible and properly positioned, supporting a seamless and fast navigation experience while clearly distinguishing between user and administrator.

Technologies Used:

I used several crucial technologies in the development of this system, which were essential for constructing both the application's frontend and backend. HTML was utilised to arrange the content of the web pages, forming the foundation for navigation and forms. Using HTML5, I ensured that the sites were semantically valid and accessible, according to contemporary online standards for clarity and usability. CSS was used to style the site pages, allowing me to develop a clean and responsive layout. I utilised basic CSS and some custom styles to guarantee that the interface was both functional and visually appealing, as well as some small layout changes to improve the user experience.

On the backend, PHP was the major language for handling server-side logic. It was used to connect to the MySQL database, retrieve and display data, and handle form submissions. PDO enabled secure database connections and queries, eliminating SQL injection and offering a safe way to communicate with the database. The data was stored and managed using MySQL, a relational database management system. The three basic tables Module, User, and Question were created to manage data for students, modules, and posted questions, and their associations were preserved via foreign keys.

These technologies, when combined, allowed for the development of a fully working CRUD system in which users could add, edit, and remove questions, as well as manage users and modules, all while interacting with an organised and secure database. The combination of PHP and MySQL enabled the dynamic production of material depending on user input, making the site more interactive and responsive to the demands of students and administrators.

Legal, Social and Ethical Issues:

Legal issues:

Compliance with data protection laws: The creation of a student-based forum system involves careful consideration of data privacy and security legislation, notably the

European Union's GDPR, which controls the collection, storage, and processing of personal data. (Goldberg, Johnson and Shriver, 2024)

General Data Protection Regulation (GDPR).

The GDPR is a rule under EU law that governs how personal data is collected, processed, stored, and destroyed. Because the system will retain personal data such as student names, email addresses, and potentially other sensitive information, the application must comply with GDPR. (Franke et al., 2024)

Data minimisation: Entails collecting just the personal information required by the system. In this assignment, collecting basic information such as the student's name, email address, and the posts they write is necessary, but no further sensitive data should be requested unless indicated and justified. (Bakare et al., 2024)

Purpose Limitation: The data gathered may only be used for the expressly mentioned objectives, such as managing posts and users in the student forum. Without Consent the data should not be used for other purposes such as marketing or advertising. (Arabsorkhi and Khazaei, 2024)

Transparency: Students must understand how their data will be utilised. This includes presenting a privacy policy outlining what data is collected, how it is handled, and the student's GDPR rights. (Franke et al., 2024)

Data Access and Control: Users should be able to access, change, and remove their data. This includes giving students the option to erase their posts and accounts. The system should provide for simple administration of personal data, including allowing students to amend or delete their information on request. (Rupp and von Grafenstein, 2024)

Data Retention: Only keep data for as long as required. For example, user postings should only be stored in the system for as long as they are required for the forum's purpose. Afterwards, data should be removed or anonymised. (Bolatito Ige, Chukwurah and Adebayo, 2024)

Social issues:

Privacy Concerns: The storing and exchange of personal data in a public setting raises various privacy concerns. In the framework of this initiative, students will most likely exchange academic-related enquiries and replies, which may include sensitive schoolwork or personal academic challenges. (Hoffmann and Boulianne, 2024)

Private Data Exposure: It is also critical to safeguard students' personal information. The system should keep email addresses private until the student actively chooses to make them public. This might be accomplished by merely displaying usernames or offering the option of private messaging. (Monika Blue Kwapisz, Kohli and Prashanth Rajivan, 2024)

Data Breach Risks: The application must secure student information against breaches. This involves storing personal data securely and encrypting it before and during transmission. (Seda Özer Şanal and Büşra Çiçek, 2023)

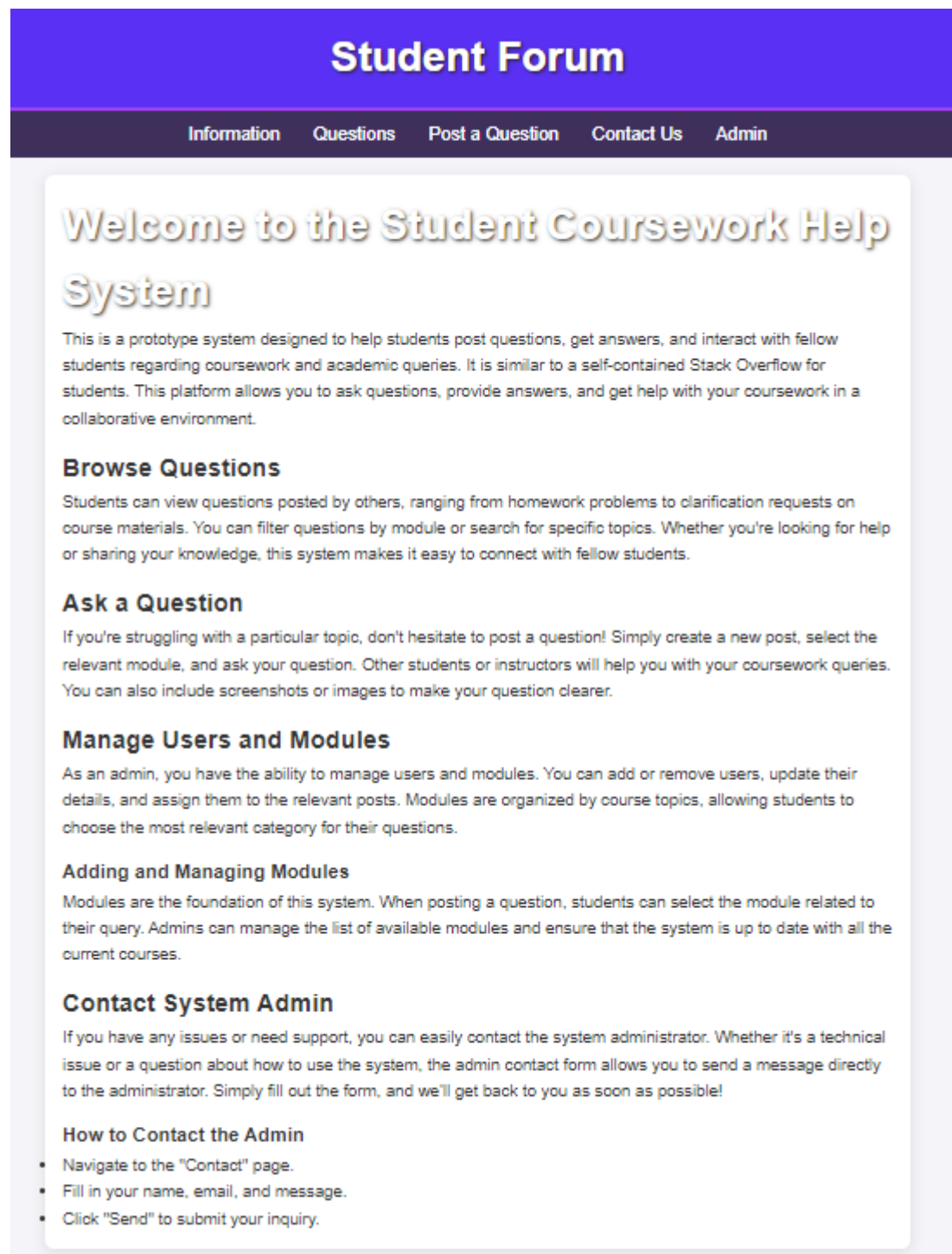
Ethical considerations:

Harassment or Bullying: The platform must guarantee that students do not face harassment, bullying, or discrimination. Monitoring options, such as reporting postings, banning people, and restricting inappropriate language, should be explored. (Blake et al., 2024)

Content Ownership: Ethical considerations may emerge around who owns the content submitted. It is critical to specify in the conditions of use whether the posts remain the intellectual property of the students or whether the platform has permission to utilise the content. (Thio, Rio Christiawan and Wagiman Wagiman, 2024)

Overview:

The system meets all of the major functional criteria, providing a seamless user experience for both students and administrators. The following is a summary of the core features, accompanied by annotated screenshots of important pages.



The Information Page introduces the platform and provides instructions on its purpose and usage. This page is very important for new users since it establishes expectations and describes how students can interact with the system. The design is straightforward and approachable, assuring clarity and readability. The page also includes connections to other important portions of the website, allowing for simple navigation.

Type your question here:

Coursework?

Select a module:

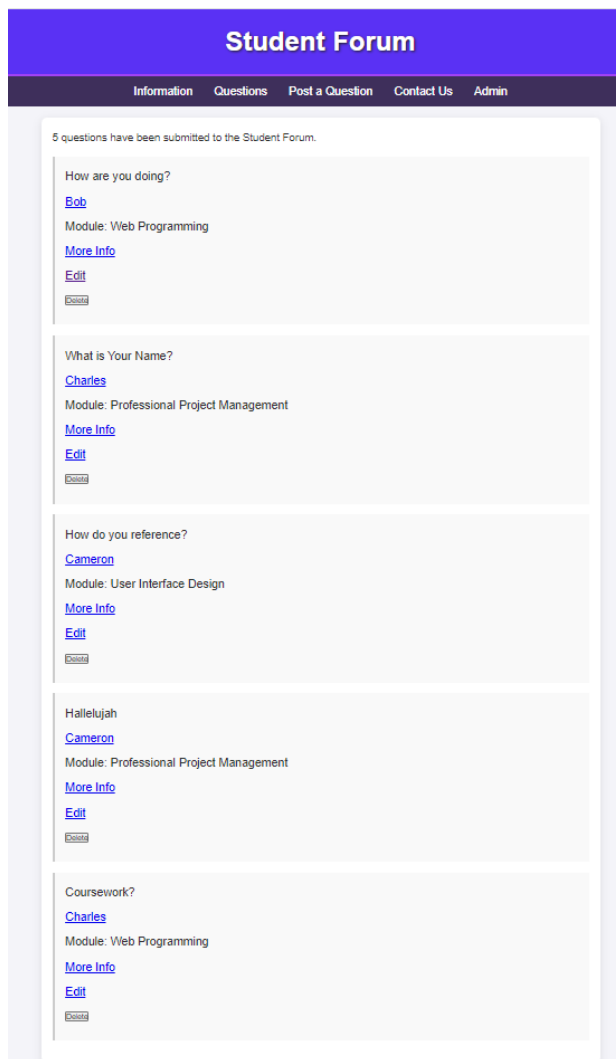
Web Programming

Select a User:

Charles

Add Question

The Add a Question Page allows users to ask their own enquiries. The form has elements for selecting the inquiry text, the user and the module. This design allows students to simply label their queries for future reference. Form validation has been introduced to prevent empty submissions and ensure all required information is delivered.



The Questions Page displays all the questions submitted by students. Each question is accompanied with the author's name, the corresponding module, and a "More Info" link, which allows viewers to access further information. The interface allows students to browse all accessible questions, making it easier to obtain pertinent information. This page meets the criteria for a public display of student-submitted questions.

Student Forum

[Information](#)[Questions](#)[Post a Question](#)[Contact Us](#)[Admin](#)

coursework test

send email



re9607h@gre.ac.uk

To:  Richard Egenti

coursework test

The Contact Us link on the public-facing homepage allows visitors to interact directly with the system administrator. This tool allows students to express questions, get assistance, or offer comments using a web-based form.

Student Forum - Add New User

Home Questions Post a Question Manage Users Manage Modules

Create a New User

Name:

Email:

Add User

Student Forum - Admin Section

Home Questions Post a Question Add User Add Module Manage Users Manage Modules

Manage Users

- Bob - Bob@gre.ac.uk [Edit](#) | [Delete](#)
- Charles - Charles@gre.ac.uk [Edit](#) | [Delete](#)
- Cameron - Cameron@gre.ac.uk [Edit](#) | [Delete](#)
- Tom - Tom@gre.ac.uk [Edit](#) | [Delete](#)

[Add New User](#)

Student Forum - Add New Module

Home Questions Post a Question Manage Users Manage Modules

Create a New Module

Module Name:

Module Code:

Add Module

Student Forum - Admin Section

Home Questions Post a Question Add User Add Module Manage Users Manage Modules

Manage Modules

- Web Programming - COMP1841 [Edit](#) | [Delete](#)
- User Interface Design - 1773 [Edit](#) | [Delete](#)
- Professional Project Management - 1770 [Edit](#) | [Delete](#)
- Computing - Comp2020 [Edit](#) | [Delete](#)

[Add New Module](#)

The Admin Section is a separate section available solely to system administrators that contains critical management tools for monitoring users and modules. The Manage and Add Users tool enables administrators to see, add, update, and remove user records. Each user's name and email address are displayed, as well as options for amending or removing their information, to ensure that user data is correct and up to date. Similarly, the Manage and Add Modules tool lets administrators browse, add, update, and remove module records. This feature is essential for keeping an organised list of coursework modules that students may use when asking queries. Together, these capabilities provide administrators with comprehensive control over the system's user and module data, ensuring the platform's seamless functioning.

Testing:

Test ID	Test Scenario	Test Description	Expected Outcome	Actual Outcome
1	View questions	Load Questions tab	Should load the view questions tab	The questions tab was loaded
2	Add a question	Insert and publish new question	A new question will be published in the questions tab	A new question was published in the questions tab
3	Edit a question	Edit the question	Questions can be edited by students and admin	Users and admin can edit posts
4	Delete a question	Delete the question	Questions can be deleted by students and admin	Users and admin can delete posts
5	Contact Us Form	Send email	Should Receive email	Email was received
6	View admin section	Load admin section	Should Load the admin section	The admin tab was loaded
7	View Users	Load manage Users tab	Should load manage users tab	The manage users tab was loaded
8	View Modules	Load manage Module tab	Should load manage module tab	The manage modules tab was loaded
9	Add new User	Insert and publish new User	Should publish new user to the system and can be seen in manage users	Admins can add new users
10	Add new Module	Insert and publish new Module	Should publish new user to the system and can be seen in manage modules	Admins can add new modules
11	Edit User	Edit User details	Should be able to edit user details	Admins can edit users in

				the manage users tab
12	Edit Module	Edit Module Details	Should be able to edit module details	Admins can edit modules in the manage modules tab
13	Delete User	Delete Users	Should delete User	Admins can delete users in the manage users tab
14	Delete Module	Delete Modules	Should delete Module	Admins can delete modules in the manage modules tab

Conclusions and future recommendations:

The system I created fits the basic needs of allowing students to publish, read, edit, and remove questions, as well as manage users and modules in the admin area. The public-facing website is straightforward and practical, with a clean design that makes navigating easy. The admin area gives you a lot of control over the data, allowing you to modify users and modules, and the "Contact Us" form lets students contact the admin.

If I had more time, I would prioritise increasing the system's functionality to increase user experience and security. This involves implementing a login system with password encryption and managing lost passwords. A more personalised strategy, such as tracking people without using a dropdown option, might simplify interactions. I would also focus on upgrading front-end design by implementing contemporary CSS frameworks for greater responsiveness and aesthetics. Furthermore, rigorous client- and server-side validation would improve the system's robustness. These enhancements would not only increase functionality, but also make the system more scalable and user-friendly in the long run.

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